

SUBJECT

300-passenger twin-aisle airliner, planform study

OBJECTIVE	Optimize the lift-to-drag ratio of a twin-aisle airliner carrying 300 passengers over a 6000 km range, with take-off at 85 m/s and landing at 72 m/s. Search the wing design space, mark any infeasible designs, and report the best feasible L/D with its envelope.
SOLVER & MODEL	Conceptual drag-polar sizing model
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RESULT

CONCEPTUAL MODEL

Best feasible cruise L/D

18.5 ± 0.5

95% confidence interval

Basis: drag-polar sizing model; no solve behind the magnitude

Winning wingspan 64 m, AR 13.7 range 9838 km

UNCERTAINTY - ASME V&V 20 THREE-CHANNEL

CHANNEL	VALUE	STATE
input	0.46	quantified
Monte-Carlo over the stated payload-mass and non-wing-drag spreads; requirements are held as exact specification, and the remaining sizing constants (SFC, fuel fraction, cruise altitude) are fixed		
numerical	—	not quantified
the design grid is discrete, so the true optimum lies between grid points		
model	—	not quantified
drag-polar sizing model; a solve would set the magnitude		

PROVENANCE - EVIDENCE-BUNDLE SEAL (SHA-256)

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Mission aircraft-optimization - Certificate C-2026-7080